

General form of the cosine function: _____

Just like graphing sine functions, we are going to use amplitude, period, phase shift, and vertical shift to graph transformed functions from the parent function of $y = \cos x$.

We are still going to graph five points, but cosine will start in a different place.... Where is that???

Example 1: Graph one period of each of the following sine functions. Make sure to clearly label your axes. Show five points plotted and list all appropriate characteristics.

a. $f(x) = 6\cos\left(\frac{1}{2}x - \pi\right)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

b. $y = 5\cos(2x + \pi) - 3$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

c. $y = -\cos\left(\frac{x+\pi}{2}\right) + 4$

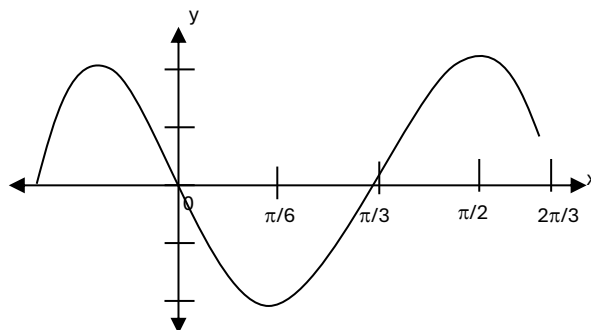
- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

Example 2: Fill out the characteristics of the function. Then, write an equation based on the given starting points.

a. Starting at $x = \frac{\pi}{6}$

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



b. Starting at $x = \frac{\pi}{2}$

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____

c. Fill out the characteristics. Then, write a cosine equation to represent the function graphed.

Amplitude _____ Period _____

Phase shift _____ Vertical shift _____

